

Project week 1: Brain Age Prediction

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Why this project?

A different kind of project

- This was originally a project from the Advanced Machine Learning course.
- Unlike the usual projects, this one gives you less guidance and leaves you more on your own.
- The goal is to use Claude to help you build a full pipeline and make the project more practical.
- There are many possible solutions. Use Claude to improve yours as much as possible and climb the leaderboard against the other teams.

Why this project? From a philosophical perspective

What this week really simulates

- This is a simulation of the real world, where you compete against others on the market for the best model.
- Having Claude does not fully solve the problem: everyone has access to it, so the baseline simply rises for all.
- The edge does not come from the tool, it comes from how you use it: your ideas, your judgment, and how you iterate.

Task: Predict a person's age from brain image data



Outline

1. Data modality: MRI
2. Task introduction
3. Dataset specifications
4. Task specifications
5. Evaluations and rules
6. Exemplar solution

Magnetic Resonance Imaging (MRI)

- Key technology in medical imaging: a magnetic field and radio waves produce detailed **3D images** of the anatomy
- **Non-invasive and radiation-free**: safe for sensitive organs like the brain
- High resolution, but expensive and **unevenly available**:
 - Switzerland: 45.0 units per million inhabitants (2016)
 - West Africa: 0.22 units per million inhabitants (2018)



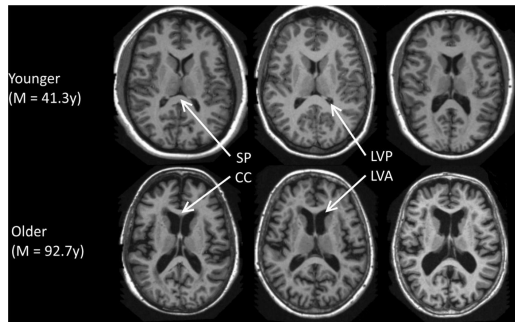
<https://data.oecd.org/healthqt/magnetic-resonance-imaging-mri-units.htm>

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Aging Effect on Brain MRI

- The brain undergoes profound age-related neuroanatomical changes during the aging process.
- The global grey matter volume decreased with age.
- Grey matter contains most of the brain's neuronal cell bodies.



Davis, Nick J. "Brain stimulation for cognitive enhancement in the older person: State of the art and future directions." *Journal of Cognitive Enhancement* 1.3 (2017): 337-344.

Brain Age Estimation: Why it is interesting?

- Aging does not affect people uniformly!
- Individual rates of aging are shaped by interactions between environmental, genetic, and epigenetic factors.
- Studies base on brain MRI show that there is a relation between accelerated aging and accelerated brain atrophy.
- It could improve early diagnosis and risk-assessments for age-associated neurodegenerative and neuropsychiatric diseases at a subject level:
 - Alzheimer, Parkinson, Huntington, etc.

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MRI Processing

Raw brain scans are difficult to handle

- 3D brain scans are $\sim 200 \times 200 \times 200 \sim 10^7$ voxels (3D voxel $\sim 1 \text{ mm}^3$)
- 3D structure + individual brain shapes $\rightarrow$ difficult to recognize disease patterns
- Data is scarce:

	ImageNet	MRI data set for Task 1
Image size	$224 \times 224 \times 3$	$\sim 200 \times 200 \times 200$
Data set size	1.2 million	$\sim 2\text{k}$

MRI Feature Extraction

We are using 832 anatomical features for this project

- Informative features derived from image data (with FreeSurfer)
- No need to process big images (6 GB) → csv sheet (3 MB)
- No need for image analysis
- Meaningful features are extracted using domain knowledge
 - e.g. cortex volume, left/right hemisphere surface area, white/gray matter volume etc.
- Information loss

Step 1: Download the data from Kaggle

Get the dataset from the competition page

- Open the competition on Kaggle:
<https://www.kaggle.com/t/40276fd0e0c54a42bcc046758201dbd1> 
- Go to the **Data** tab
- On the right, click *Download All*
- Unzip the downloaded archive
 - You should get `X_train.csv`, `y_train.csv`, `X_test.csv` and `sample.csv`

Step 2: Open the folder in VS Code

Set up your project workspace

- Create a new folder for the project (for example `brain_age`)
- Move the downloaded CSV files into that folder
- Open **VS Code**
- Go to **File** → **Open Folder** and select your project folder

Step 3: Use Claude

Let Claude help you build the solution

- Open the integrated terminal (**Terminal** → **New Terminal**)
- Start Claude Code by typing `claude`
- Ask Claude to explore the data and build the model
- Iterate directly in the chat

Share your project with GitHub

Collaborate with your teammates

- To share the project with your team, use **GitHub**
- Create a repository on github.com 
- Push your project folder to the repository:

```
git init
git add .
git commit -m "initial commit"
git remote add origin <your-repo-url>
git push -u origin main
```

- Your teammates can then clone the repository and work together
- Claude can help you with all the git commands

New to GitHub? This short video explains how it works: youtube.com/watch?v=h2a3Kw-l_Ec 

File Description

We provide the following files:

- **X_train.csv**, **y_train.csv**: the training set, including the features and labels
- **X_test.csv**: the testing set (make predictions based on this file)
- **sample.csv**: a sample submission file in the correct format

What the data looks like

One row per subject: features extracted from the brain MRI, and the age to predict

id	832 features (X_train.csv)				age (y_train.csv)
	x0	x1	x2	...	y
0	14168.82	10514.38	3316.15	...	71
1	17757.04	9814.20	4101.02	...	44
2	14226.66	11029.64	3714.51	...	83
Test set (X_test.csv): same features, age unknown					
0	15320.11	10098.73	3550.98	...	?
1	16011.45	10877.02	3902.66	...	?

Your model learns the relation features $\rightarrow$ age on the training set, then predicts the ? for every subject in the test set.

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Task 1: Age Prediction

We have modified the derived input data in three ways:

- Irrelevant features
- Outliers
- Perturbations (e.g. missing values, etc)

id	x0	x1	x2	x3	x4	...
0	14168.82	10514.38	3316.15	94230.7	102.3866	...
1	17757.04		4101.016	92559.53	99.85517	...
2	14226.66	11029.64		124055.6	100.5425	...

Subtask 0: Filling Missing Values

Background

There are missing values in the data

- Originally they are set to NaN
- Most methods cannot handle NaNs automatically
- Different possible strategies to impute missing values: mean, median, most frequent etc.

Task requirements

We require that students fill missing values in the training and the test set

Subtask 1: Outlier Detection

Background

In the training set, there exists outliers

- If the resulting model is not robust enough, it may be sensitive to the outliers
- Solution: outlier removal

Task requirements

We require that students build an outlier detection model to make classification for samples in the training set i.e. whether they are outliers.

Subtask 2: Feature Selection

Background

To make the task a bit more challenging, we added some manual features to the FreeSurfer-processed dataset.

- Feature selection is needed

Advantages:

- Simplifies the models to make them interpretable
- Leads to shorter training times
- Better generalization by reducing overfitting

Task requirements

We require that students use feature selection methods to label the features as selected features and unselected features.

Here, unselected features includes irrelevant features and redundant features.

Main Task: Age Prediction

Background

After primary preprocessing and dimensionality reduction, now we finally arrive at the regression task.

Task requirements

We require that students use suitable regression methods to predict the age of a person from brain data.

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Evaluation Metrics

Coefficient of Determination R^2

is the proportion of the variance in the dependent variable that is predictable from the independent variable.

$$R^2 := 1 - \frac{SS_{res}}{SS_{tot}}$$
$$SS_{tot} := \sum_i (y_i - \bar{y})^2$$
$$SS_{res} := \sum_i (f_i - y_i)^2$$

- Varies between 1 (best) and $-\infty$;
 $SS_{tot} = SS_{res}$ then $R^2 = 0$
- R^2 is a scaled version of MSE
e.g. R^2 is invariant to scaling y , unlike MSE

How to compute it in Python:

```
from sklearn.metrics import r2_score  
  
score = r2_score(y_true, y_pred)
```

Submission to Kaggle

Kaggle link: <https://www.kaggle.com/t/40276fd0e0c54a42bcc046758201dbd1> 

- Team names must be alphanumeric (A-Z, a-z, 0-9).
- Deadline: 08/11/2026 11:59 PM
- Including public/private leaderboard:
 - Public leaderboard is available
 - Private leaderboard will open after the deadline
- Public baseline for passing the project: 0.5
- **Project defense**

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Basic Examples: Data loading

```
import os
import numpy as np
import pandas as pd

DATA_DIR = "eth-fdd-competition" # folder containing the csv files

X_train_df = pd.read_csv(os.path.join(DATA_DIR, "X_train.csv"), skiprows=1, header=
    None)
y_train_df = pd.read_csv(os.path.join(DATA_DIR, "y_train.csv"), skiprows=1, header=
    None)
X_test_df = pd.read_csv(os.path.join(DATA_DIR, "X_test.csv"), skiprows=1, header=
    None)

X_train = X_train_df.values[:, 1:]
y_train = y_train_df.values[:, 1:]
X_test = X_test_df.values[:, 1:]

print(X_train.shape, y_train.shape, X_test.shape)
```

Basic Examples: Splitting and Imputing

```
from sklearn.model_selection import train_test_split

# Randomly split the data into training and validation sets with 80-20 ratio
X_train, X_val, y_train, y_val = train_test_split(X_train, y_train, test_size=0.2,
                                                random_state=42)

# Impute missing values with mean of each column
X_mean = np.nanmean(X_train, axis=0, keepdims=True)
X_train = np.where(np.isnan(X_train), X_mean, X_train)
X_val = np.where(np.isnan(X_val), X_mean, X_val)
X_test = np.where(np.isnan(X_test), X_mean, X_test)
^^I
```

Basic Examples: Feature Selection

```
from sklearn.feature_selection import SelectKBest, mutual_info_regression

# Select top 100 features with highest mutual information
selection = SelectKBest(mutual_info_regression, k=100).fit(X_train, y_train.ravel())
)
X_train = selection.transform(X_train)
X_val = selection.transform(X_val)
X_test = selection.transform(X_test)
^^I
```

Basic Examples: Training and Validating

```
from sklearn.linear_model import LinearRegression
from sklearn.metrics import r2_score

# Train a linear regression model
regressor = LinearRegression()
regressor.fit(X_train, y_train)

y_train_pred = regressor.predict(X_train)
y_val_pred = regressor.predict(X_val)

# Evaluate the model on training and validation sets
train_score = r2_score(y_train, y_train_pred)
val_score = r2_score(y_val, y_val_pred)

print(train_score, val_score)
^^I
```

Basic Examples: Export and Submit

```
# Predict on test set
y_test_pred = regressor.predict(X_test)

# Save predictions to submission file with the given format (id, y)
OUTPUT_DIR = "output"
os.makedirs(OUTPUT_DIR, exist_ok=True)
table = pd.DataFrame({"id": np.arange(0, y_test_pred.shape[0]), "y": y_test_pred.
    flatten()})
table.to_csv(os.path.join(OUTPUT_DIR, "submission.csv"), index=False)
^^I
```

In Kaggle: *New submission* → choose `submission.csv` → add a description → *Add submission*.

Q&A